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CONTINUUM 4D SU(N) YANG–MILLS WITH A MASS GAP

Main result (conditional): under stated renormalization-group, strong-coupling endpoint and continuum hypotheses — above all, uniform error control through the crossover ($\theta > 0$) — the limiting gauge-invariant observable family satisfies the Osterwalder–Schrader axioms, and its reconstructed Hamiltonian has a mass gap $m > 0$. Nontriviality and identification with Yang–Mills remain separate obligations; no solution of the Millennium Problem is claimed.

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Abstract

This paper presents a conditional research programme for matching weak-coupling lattice Yang–Mills theory to a strong-coupling endpoint. The six workstreams are retained; their propositions describe tasks rather than established theorems. Lemmas 4 and 5 give algebraic implications under unproved spectral-comparison and endpoint hypotheses, with the stopping scale measured against a stipulated two-loop scale. Lemma 2 supplies a finite-volume Hamiltonian–Euclidean bridge, and Lemma 3 records classical exact two-dimensional heat-kernel identities with explicit domain restrictions. The continuum theorem requires strengthened hypotheses for a full generating observable algebra, distributional convergence, Euclidean covariance and reconstruction. Nontriviality and identification with Yang–Mills remain separate obligations. No solution of the Millennium Problem, and no new unconditional four-dimensional construction, is established.

Provenance

Provenance B2 in the companion paper "Toward the Weak-Coupling Continuum Limit of SU(N) Lattice Yang–Mills" denotes the workstream diagram and the six workstreams. The flowchart below is that diagram. The table maps each box to its Proposition, the workstream whose statement it uses, and the gate that becomes the Lemma's hypothesis.

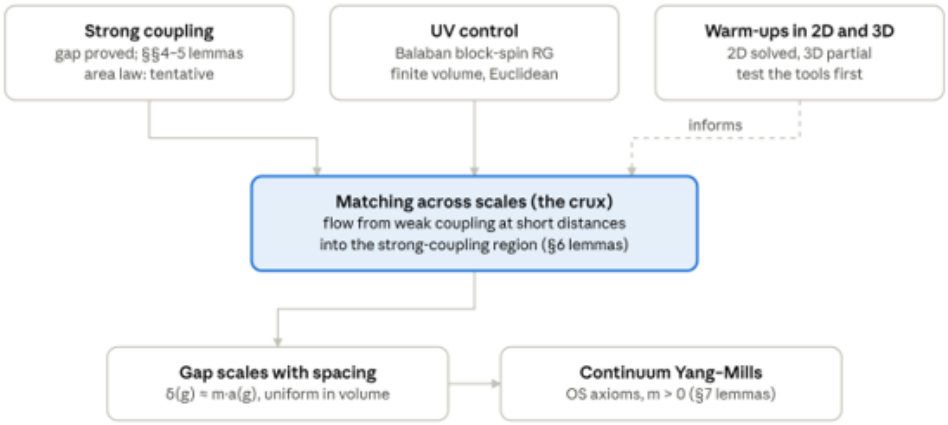


Figure 1. The B2 flowchart: three inputs, one crux, two outcomes.

Box	Proposition	Workstream	Gate (hypothesis of the Lemma)
Strong coupling	Proposition 1	1	$\mathfrak{G}_1$: certified Yarotsky constants and a uniform link-embedding estimate
UV control	Proposition 2	2	$\mathfrak{G}_2$: reflection positivity of the Euclidean measures
Warm-ups in 2D and 3D	Proposition 3	3	$\mathfrak{G}_3$: the block-spin step maps the model class into itself on b-block domains
Matching across scales (the crux)	Proposition 4	5	$\mathfrak{G}_4$: uniform spectral error control through the crossover, $\theta > 0$
Gap scales with spacing	Proposition 5	4	$\mathfrak{G}_5$: finite RG stopping time
Continuum Yang–Mills	Proposition 6 and Theorem	6	C_1–C_4

The transfer-matrix bridge identifies finite-volume temporal correlations. Applying Euclidean renormalization results still requires control of the anisotropic action and of the time-continuum limit. Workstream 4 requires two-sided spectral comparison and scale control: absence of a gap closing alone does not give $\delta(g) \asymp a(g)$, nor does it exclude every thermodynamic phase transition.

0 Notation and standing hypotheses

Model. $G = SU(N)$, $N \geq 2$, metric $-\mathrm{Tr}(XY)$, $c_N = (N^2-1)/N$. The universal two-loop coefficients are $b_0 = 11N/(48\pi^2)$ and $b_1 = 17N^2/(384\pi^4)$. Λ_L is the open cubic box $\{0,\dots,L\}^3$. The Kogut–Susskind Hamiltonian is

$$H_L(g) = (g^2/2)\,\Sigma_e\,(-\Delta_e) + g^{-2}\,\Sigma_p\,[N - \mathrm{ReTr}\,\rho(U_p)],$$

with unique ground state Ω_L, energy $E_{0,L}$, $P_{0,L} = |\Omega_L\rangle\langle\Omega_L|$, and $\delta_L(g)$ its gap on physical states in lattice units. $\lambda = g^{-4}$, $\langle\cdot\rangle_L = \langle\Omega_L, \cdot\,\Omega_L\rangle$, $W_L(C) = N^{-1}\,\mathrm{Tr}\,\rho(\Pi_{e\in C}\,U_e)$, and $\mathscr{A}_L$ denotes bounded gauge-invariant multiplication operators. Set

$$a(g) := \Lambda^{-1}\,(b_0 g^2)^{-b_1/(2b_0^2)}\,\exp(-1/(2b_0 g^2)).$$

This is a stipulated two-loop scale, not a proved spacing. Including the two-loop factor matters: relative to the one-loop scale, two-loop running shifts the RG stopping scale by an unbounded multiple of $\ln(1/g^2)$, whereas relative to $a(g)$ the remaining perturbative corrections are factors $1 + O(g^2)$.

Strong-coupling class. $\mathscr{Y}(\beta)$ consists of normalized product Hamiltonians with on-site gap at least one and finite-range bounded perturbations satisfying all hypotheses of [3, Theorem 1], with range $\Lambda_0 = \{0,1,2\}^3$ and $\sup_x \|\varphi_x\| \leq \beta$. Choose certified smallness constants $c_1, c_2 > 0$ with gap bound $1 - c_2\beta$, and set $\beta^* = \min\{c_1/2, (2c_2)^{-1}\}$, $g_0(N) = (6N/(c_N\beta^*))^{1/4}$. Uniformity for the link embedding and all boundary terms is required; [1] is an unverified companion input.

Expansion hypothesis $\mathfrak{E}(g, K)$. $\langle W_L(C)\rangle_L = \Sigma_{X\in\mathscr{P}(C)}\,w_X(C)$, where $\mathscr{P}(C)$ is the set of connected plaquette sets containing a plaquette that contains a link of C; each w_X is a finite sum of matrix elements $\langle F\Omega_0, W(C)F'\Omega_0\rangle$ with F, F' built from $\{\psi_p\}_{p\in X}$ and functions of $\Sigma_e(-\Delta_e)$; and $|w_X| \leq (K\lambda)^{|X|}$, uniformly in L.

Euclidean objects. For $\tau > 0$, $T_\tau = e^{-\tau V/2}\,e^{-\tau(g^2/2)K}\,e^{-\tau V/2}$, with $K = \Sigma_e(-\Delta_e)$ and $V = g^{-2}\,\Sigma_p[N - \mathrm{ReTr}\,\rho(U_p)]$. $\langle\cdot\rangle_{L,\tau}$ is the anisotropic lattice measure on $\Lambda_L \times \tau\mathbb{Z}$ with transfer matrix T_τ (heat-kernel time-like weights, Wilson space-like weights), taken in infinite temporal extent. F_t is F placed on time slice t/τ, Θ is time reflection, and $\langle\cdot\rangle^T$ is the truncated expectation.

Two-dimensional objects. $D \subset \mathbb{Z}^2$ is a finite simply connected union of plaquettes. $K_t(U) = \Sigma_\lambda\,d_\lambda\,\chi_\lambda(U)\,e^{-t\mathfrak{C}(\lambda)/2}$ is the heat kernel, and $\mu_t^D = Z^{-1}\,\Pi_{p\in D}\,K_t(U_p)\,\Pi_e\,dU_e$ with free boundary. $\mathscr{B}_b$ is the path-product block spin [2, Lemma 6.2], and D/b is the coarse region when D is a union of b-blocks.

Renormalization-group hypotheses. Fix $b \geq 2$. For $0 < g < g_0$ and admissible boxes $L \in b^{k^*}\mathbb{N}$, assume effective physical Hamiltonians $H^{(k)}$ on Λ_{L/b^k}, $H^{(0)} = H_L(g)$, with couplings g_k and gaps $\delta^{(k)}$ in scale-k units. Set $k^*(g) = \min\{k \geq 0 : g_k \geq g_0\}$, and $k^* = 0$ for $g \geq g_0$. All estimates below are uniform on these boxes.

- R: $\delta^{(k-1)} \geq b^{-1}(1 - \eta_k)\,\delta^{(k)}$ with $\eta_k \in [0, 1)$, for $1 \leq k \leq k^*$.
- R′: $\delta^{(k-1)} \leq b^{-1}(1 + \eta_k)\,\delta^{(k)}$, and $\Theta := \sup_{g,L}\,\Pi_{k\leq k^*}(1 + \eta_k)$.
- Y: $2(g_{k^*}^2\,c_N)^{-1}(H^{(k^*)} - \mathrm{const}) \in \mathscr{Y}(\beta^*)$ and $g_{k^*} \geq g_0$. Y′: $\delta^{(k^*)} \leq D$.
- AF: $k^*(g)\ln b \leq \ln(a(g_0)/a(g)) + \kappa \ln b$. AF±: $|k^*(g)\ln b - \ln(a(g_0)/a(g))| \leq \kappa \ln b$.

Continuum hypotheses ($g \downarrow 0$, $L \cdot a(g) \to \infty$ through admissible boxes; take $\tau \downarrow 0$ first). Use a renormalized generating family of local gauge-invariant fields, including tensor components, and its full polynomial algebra:

- C_1: $\exists\, m > 0$: $|\langle A\, B_x \rangle^T| \le C_{A,B} \exp(-m|x|)$ for every compactly smeared polynomial A, B, uniformly away from boundaries.
- C_2: all mixed S_n converge in tempered distributions, independently of the sequence, with $S_0 = 1$ and the field adjoint relations.
- C_3: the limiting family is $E(4)$-covariant, including translations and field indices.
- C_4: the limiting family satisfies the precise OS regularity and growth assumptions of [12].

1 Strong coupling

Proposition 1. Finish the strong-coupling side. Prove the area law (Steps one and three of the strong-coupling white paper) and extract Yarotsky's constants c_1, c_2.

Deliverable: gap and area law at strong coupling with an explicit $g_0(N)$. Gate: a numerical $g_0(N)$.

Lemma 1.

$\mathfrak{G}_1 : c_1, c_2 > 0$ certified $\wedge \sup_x \|\varphi_{2x}\| \le 6N/(g^4 c_N)$ on the link embedding, boundary terms included; β^*, $g_0(N)$ as in §0

(i) $\forall\, g \ge g_0(N), \forall\, L \in \mathbb{N}: \quad H_L(g) - E_{0,L} \ge (g^2 c_N/4)(\mathbb{1} - P_{0,L})$

(ii) $\mathfrak{E}(g, K) \wedge q := 12e\, K\, g^{-4} \le \tfrac{1}{2} \implies \forall\, L, \forall\, C \subset \Lambda_L\ (r \times s): \quad |\langle W_L(C)\rangle_L| \le M\, e^{-\sigma rs}, \quad \sigma = \tfrac{1}{2} \ln q^{-1},$ $M = 32/(e\sigma)$

Proof. Part (i) requires a uniform application of [3] to the normalized link Hamiltonian. After subtracting the constant part of the plaquette potential and rescaling by $2/(g^2 c_N)$, $\mathfrak{G}_1$ gives grouped perturbations of norm at most $6N/(g^4 c_N) \le \beta^*$. The certified endpoint estimate then gives normalized gap at least $1 - c_2\beta^* \ge \tfrac{1}{2}$, and rescaling gives $g^2 c_N/4$. The unique positive ground state is gauge invariant, so restriction to physical states preserves the lower bound. The embedding and boundary estimates cited from [1] must be supplied to make this self-contained.

For (ii), fix a rectangle C. Each w_X is a sum of matrix elements to which the center selection rule [1, Lemma 5.1] applies, so $w_X = 0$ unless X supports a $\mathbb{Z}_N$ 2-chain Σ with $\partial\Sigma = C$. By the area count [1, Proposition 5.2], $|X| \ge rs$. Every link of C has nonzero coefficient in $\partial\Sigma$, so X contains a plaquette containing a link of C; there are at most $4 \cdot 2(r + s)$ such plaquettes. In the plaquette adjacency graph (two plaquettes adjacent if they share a link) each plaquette has at most $4 \cdot 3 = 12$ neighbours, so the number of connected sets of size n containing a given plaquette is at most $(12e)^n$. Hence

$$|\langle W_L(C)\rangle_L| \le \Sigma_{n \ge rs}\, 8(r+s)(12e)^n (K\lambda)^n = 8(r+s)\, \Sigma_{n \ge rs}\, q^n \le 16(r+s)\, q^{rs},$$

using $q \le \tfrac{1}{2}$. Since $r + s \le 2rs$ and $x\, q^{x/2} \le 2/(e \ln q^{-1})$ for $x \ge 0$, we get $16(r+s)q^{rs} \le 32\, rs\, q^{rs/2} \cdot q^{rs/2} \le (64/(e \ln q^{-1}))\, q^{rs/2} = M\, e^{-\sigma rs}$. All bounds are uniform in L. ∎

The gap estimate is conditional on the certified embedding input; it is not independently verified here. The area-law estimate also requires the open expansion $\mathfrak{E}(g, K)$ and its center-support selection property.

2 UV control

Proposition 2. Bridge Hamiltonian and Euclidean. Show the Hamiltonian is the time-continuum limit of an anisotropic Euclidean lattice theory, using the transfer matrix and reflection positivity.

Deliverable: a finite-volume transfer-matrix bridge and a spectral characterization of temporal decay. Gate: compatibility with any Euclidean estimates to be transferred must be proved.

Lemma 2.

$\mathfrak{G}_2 : \forall\, \tau > 0, \forall\, F \in \mathcal{A}_L^{+} :\ \langle \Theta(\bar{F})\, F\rangle_{L,\tau} \geq 0,\quad \mathcal{A}_L^{+} :=$ positive-time gauge-invariant cylinder algebra

(i) $\forall\, F, G \in \mathcal{A}_L, \forall\, t \geq 0 :\ \lim_{\tau \downarrow 0,\, t/\tau \in \mathbb{N}} \langle F_0\, G_t\rangle_{L,\tau} = \langle \Omega_L, F\, e^{-t(H_L - E_{0,L})}\, G\, \Omega_L\rangle$

(ii) $\delta_L(g) = \inf_{F \in \mathcal{A}_L} \operatorname{liminf}_{t\to\infty} \left(-t^{-1} \ln |\langle \bar{F}\, e^{-t(H_L - E_{0,L})}\, F\rangle^{T}_{L}| \right)$

Proof. *Gate.* $e^{-\tau(g^2/2)K}$ is a positive operator, so $T_\tau = e^{-\tau V/2}\, e^{-\tau(g^2/2)K}\, e^{-\tau V/2}$ is positive and self-adjoint. For F supported in positive times, $\langle \Theta(\bar{F})F\rangle_{L,\tau}$ is, up to a positive normalization, a quantity of the form $\langle \psi_F, T_\tau^{\,j}\, \psi_F\rangle$ with $j \geq 0$, which is nonnegative. This is the Osterwalder–Seiler transfer-matrix argument [8].

(i) At fixed L and g, K is elliptic on a compact product of groups and V is smooth and bounded. The symmetric product formula converges in norm for each fixed positive time [9]; no uniformity in L or g is asserted. Strict positivity of the heat kernel gives a simple top eigenvalue of T_τ. Norm convergence of a fixed-time power $T_\tau^{\,n}$, with $n\tau$ tending to a positive reference time, gives convergence of its isolated ground projection and normalized top eigenvalue, and the ground vector is also that of T_τ. Inserting bounded F, G and using the product formula at time t gives (i), with $t = 0$ obtained directly from ground-state convergence.

(ii) Let μ_F be the spectral measure of $A_L = H_L - E_{0,L}$ for $Q_L F \Omega_L$. Then $\langle \bar{F}\, e^{-tA_L}\, F\rangle^{T} = \int_{(0,\infty)} e^{-tE}\, d\mu_F(E)$, whose exponential decay rate is $\inf \operatorname{supp} \mu_F \geq \delta_L(g)$. Since $\Omega_L > 0$, the vectors $F\Omega_L$ with $F \in \mathcal{A}_L$ are dense in the physical space, so for every $\varepsilon > 0$ some F has $\mu_F([\delta_L, \delta_L + \varepsilon)) > 0$. Its rate is then at most $\delta_L + \varepsilon$, and the infimum equals $\delta_L(g)$. ∎

Lemma 2 holds at fixed finite volume. It does not transfer Balaban's small-field estimates [5], or the isotropic strong-coupling estimates of Shen–Zhu–Zhu [6], to this Hamiltonian without further action comparison and uniform time-limit estimates. Strong coupling in [6] need not be uniform as $\tau \downarrow 0$.

3 Warm-ups in 2D and 3D

Proposition 3. Warm up in lower dimensions. Test the selection rule, the curvature criterion and the matching idea where ultraviolet problems are milder (3D Euclidean, i.e. the 2+1 Hamiltonian model), checked against exact 2D results.

Deliverable: a worked matching argument in a setting with controllable renormalization. Gate: the method survives one full scale-matching in a simpler theory.

Lemma 3.

$\mathfrak{G}_3 : \mathscr{B}_{b^*}\, \mathscr{K} \subset \mathscr{K}, \quad \mathscr{K} := \{\mu_t^D : t > 0,\ D \text{ a union of b-blocks}\};$ identities of measures on the gauge quotient

(i) $\mathscr{B}_{b^*}\, \mu_t^D = \mu_{b^2 t}^{D/b}$

(ii) $\forall\, C \subset D$ simple, contractible : $\langle W(C)\rangle_{\mu_t^D} = \exp(-t\, c_N\, |A(C)|/2)$

(iii) $k := \max\{0, \lceil \ln(t_0/t)/\ln(b^2)\rceil\}$; D a union of b^k-blocks $\Longrightarrow$ $\mathscr{B}_{b^*}^{\,k}\, \mu_t^D = \mu_{b^{2k} t}^{D/b^k} \wedge b^{2k} t \geq t_0$

Proof. Choose a planar peeling order and compatible rooted lasso generators, obtained by successively removing boundary faces, rather than an arbitrary choice of plaquette lassos. After tree gauge fixing, each removed face eliminates one independent link by a Haar-preserving group translation. Iterating gives the product density $\Pi_p K_t(L_p) dL_p$ for the lasso variables. Residual gauge invariance acts by simultaneous conjugation. This establishes the independent-face representation used below [10].

(i) Integrate the internal links of each coarse plaquette successively. At every gluing step, centrality and Haar invariance give the heat-kernel convolution identity $K_s * K_t = K_{s+t}$. A block of b^2 plaquettes therefore has boundary weight $K_{b^2 t}$. Disjoint blocks can be integrated independently, and the retained coarse links carry Haar measure. This gives the claimed quotient measure, hence all gauge-invariant expectations. Arbitrary variable-dependent conjugations need not preserve independence; the integration argument avoids that assumption.

(ii) The holonomy of C is a product of $|A(C)|$ independent conjugated lassos. Since $\int K_t(U)\, \chi_\lambda(UV)\, dU = e^{-tC(\lambda)/2}\, \chi_\lambda(V)$, iterating over the $|A(C)|$ lassos with λ the fundamental representation ($C = c_N$, $\chi(\mathbb{1}) = N$) gives $\langle W(C)\rangle = \exp(-t\, c_N\, |A(C)|/2)$.

(iii) Iterate (i) k times when the stated block divisibility holds; $k = 0$ means the identity map. ∎

In two dimensions these classical heat-kernel identities give exact blocking and an area law at every $t > 0$ [10, 11]. Reaching t_0 requires a sufficiently large divisible domain, or the infinite plane. This does not establish the four-dimensional Hamiltonian hypotheses R, Y or AF. The 3D and 2+1 warm-up remains open.

4 Matching across scales (the crux)

Proposition 4. Match across scales (the crux). Run a Balaban-type block-spin RG from weak coupling at short distances until the effective coupling exceeds $g_0(N)$, then apply the strong-coupling gap to the effective theory.

Deliverable: $\delta(g) \gtrsim a(g)$ for the stipulated scale. Gate: uniform spectral error control through the crossover, endpoint control and a quantitative stopping-scale bound.

Lemma 4.

$$\mathfrak{G}_4 : \theta := \inf_{g\in(0,g_0),\, L} \prod_{k=1}^{k^*(g)} (1 - \eta_k) > 0$$

$$\mathsf{R} \wedge \mathsf{Y} \wedge \mathsf{AF} \wedge \mathfrak{G}_4 \implies \forall\, g \in (0, g_0), \forall\, L \in b^{k^*(g)}\mathbb{N} : \quad \delta_L(g) \geq C_4\, a(g),$$

$$C_4 := \theta \cdot (g_0^2\, c_N/4) \cdot b^{-\kappa} / a(g_0)$$

Proof. Fix $g \in (0, g_0)$ and $L \in b^{k^*}\mathbb{N}$, so that every Λ_{L/b^k} with $k \leq k^*$ is a box.

1. *Strong-coupling endpoint.* By Y the normalized Hamiltonian $2(g_{k^*}^2 c_N)^{-1}(H^{(k^*)} - \text{const})$ lies in $\mathscr{Y}(\beta^*)$. The certified estimate gives normalized gap at least $1 - c_2\beta^* \geq \tfrac{1}{2}$, so $\delta^{(k^*)} \geq g_{k^*}^2 c_N/4 \geq g_0^2 c_N/4$.
2. *Descent.* Iterating R from $k = k^*$ down to $k = 1$ gives $\delta^{(0)} \geq b^{-k^*} \prod_{k\leq k^*}(1 - \eta_k)\, \delta^{(k^*)} \geq \theta\, b^{-k^*}\, g_0^2 c_N/4$, using $\mathfrak{G}_4$.
3. *Stopping scale.* By AF, $-k^* \ln b \geq -\ln(a(g_0)/a(g)) - \kappa \ln b$, that is, $b^{-k^*} \geq b^{-\kappa}\, a(g)/a(g_0)$.
4. Since $\delta^{(0)} = \delta_L(g)$, combining steps 2 and 3 gives $\delta_L(g) \geq C_4\, a(g)$. Nothing depends on L beyond divisibility, so the bound is uniform on admissible boxes. ∎

Lemma 4 is a conditional spectral comparison on admissible boxes. A sufficient criterion for $\theta > 0$ is a uniform bound on $\Sigma_k -\ln(1 - \eta_k)$. Constructing the effective Hamiltonians, controlling all induced interactions and proving these bounds remain open. AF is stated relative to the two-loop scale $a(g)$: measured against the one-loop scale, a bounded κ would fail, because two-loop running shifts the stopping scale by an unbounded multiple of $\ln(1/g^2)$. Relative to $a(g)$, bounded κ is consistent with perturbative running, but it is not proved. Lemma 3 is an analogy, not a verification of these four-dimensional hypotheses.

5 Gap scales with spacing

Proposition 5. Control the gap at intermediate coupling. Seek positivity of $\delta_L(g)$ at every fixed $g > 0$, uniformly on admissible volumes, together with two-sided scale bounds.

Deliverable: conditional exclusion of gap closing on the stated trajectory. Gate: finite RG stopping time, uniform comparisons and a certified strong-coupling endpoint.

Lemma 5.

$\mathfrak{G}_5 : \forall\, g > 0 : \quad k^*(g) < \infty$

$\mathfrak{G}_1 \wedge \mathsf{R} \wedge \mathsf{R}' \wedge \mathsf{Y} \wedge \mathsf{Y}' \wedge \mathsf{AF}\pm \wedge \mathfrak{G}_4 \wedge \mathfrak{G}_5 \wedge \bar{\Theta} < \infty \implies$

(i) $\forall\, g > 0 : \quad \inf\{\delta_L(g) : L \in b^{k^*(g)}\mathbb{N}\} > 0$

(ii) $0 < C_4 \le \liminf_{g\downarrow 0} \inf_L \delta_L(g)/a(g) \le \limsup_{g\downarrow 0} \sup_L \delta_L(g)/a(g) \le C_5 < \infty, \quad L \in b^{k^*(g)}\mathbb{N},$

$C_5 := \bar{\Theta} \cdot D \cdot b^{\kappa} / a(g_0)$

Proof. (i) At $g \ge g_0$ use Lemma 1(i), including its certified embedding input. For $0 < g < g_0$ apply Lemma 4; $\mathsf{AF}\pm$ implies AF. This yields positivity uniformly on admissible boxes. Extension to all L requires an additional boundary comparison estimate, which is not assumed here.

(ii) The lower bound is Lemma 4 divided by $a(g)$. For the upper bound, iterate R' from k^* down to 1: $\delta^{(0)} \le b^{-k^*} \Pi(1 + \eta_k)\, \delta^{(k^*)} \le \Theta D\, b^{-k^*}$, using Y'. By the lower half of $\mathsf{AF}\pm$, $k^* \ln b \ge \ln(a(g_0)/a(g)) - \kappa \ln b$, so $b^{-k^*} \le b^{\kappa}\, a(g)/a(g_0)$. Hence $\delta_L(g) \le C_5\, a(g)$. Both constants are independent of g and L. ∎

Part (ii) bounds δ_L/a on the stipulated two-loop trajectory; it proves neither convergence nor a nonzero continuum spectral weight. Positivity of a gap does not by itself exclude a first-order bulk transition or prove analyticity of the free energy. Physical spacing and the coupling normalization must still be matched to renormalized observables.

6 Continuum Yang–Mills

Proposition 6. Take the continuum and infinite-volume limits. Prove existence and uniqueness of the limit, the Osterwalder–Schrader axioms and $m > 0$.

Deliverable: conditional OS reconstruction in the generated observable sector; Yang–Mills identification is an additional task.

Theorem (conditional continuum reconstruction with a mass gap). Assume the strengthened continuum hypotheses C_1–C_4 of Section 0 for the full generating field algebra, obtained from reflection-positive lattice expectations with reflection-compatible renormalization. Then the limiting family satisfies OS0–OS4 and reconstructs a theory $(\mathscr{H}, H, \Omega)$ in its vacuum-generated sector with unique vacuum and

$$\operatorname{spec}(H) \subset \{0\} \cup [m, \infty), \quad m > 0,$$

where m is a lower bound, not necessarily the exact spectral threshold. Lemmas 2 and 5 separately bound lattice temporal rates on admissible boxes under their RG and certified endpoint assumptions.

Proof.

1. *Existence and uniqueness.* By C_2.
2. *OS0 (temperedness and growth).* By C_4.
3. *OS1 (Euclidean covariance).* By the strengthened C_3. Finite open boxes are not translation invariant, so $E(4)$ covariance must be established in the joint limit, including removal of boundary effects. Temperedness alone gives no pointwise continuity of Schwinger distributions.
4. *OS2 (reflection positivity).* Use $\mathfrak{S}_2$ for arbitrary finite polynomials supported at positive times. Reflection-compatible smearing and renormalization, followed by the full distributional convergence in C_2, preserve each finite nonnegative quadratic form.
5. *OS3 (symmetry).* The observables are commuting multiplication operators, so the S_n are symmetric.
6. *OS4 (clustering).* The strengthened C_1 applies to every pair of polynomial clusters and passes to the limit. This gives clustering and uniqueness of the vacuum in the generated sector.
7. *Reconstruction.* By the Osterwalder–Schrader reconstruction theorem [12], OS0–OS4 yield a Hilbert space $\mathscr{H}$, a self-adjoint $H \geq 0$ and a unique vacuum Ω.
8. *Mass gap.* For any positive-time polynomial A, let $\psi = (A - \langle A \rangle)\Omega$ in the reconstructed space. Reflection positivity and time translation identify $\langle \psi, e^{-tH}\psi \rangle$ with a connected correlation of two polynomial clusters. By C_1 it is bounded by $C_A\, e^{-mt}$ at large t. If its positive spectral measure assigned weight $\eta > 0$ to $[0, m - \varepsilon]$, this correlation would be at least $\eta\, e^{-(m-\varepsilon)t}$, a contradiction. Polynomial vectors are dense by reconstruction, so the spectral projection on $(0, m)$ vanishes, and clustering removes extra zero-energy vectors. Scalar two-point decay alone would not justify this step.
9. *Scope.* Nontriviality, renormalized local curvature fields and their Yang–Mills dynamics are not consequences of OS reconstruction. A nonzero connected observable, and identification of the limiting field algebra and coupling scheme, must be proved separately. Lattice gap bounds alone neither identify the continuum theory nor determine its exact mass. ∎

7 Status of the hypotheses

The Theorem is a reduction: it is exactly as strong as its hypotheses. This table lists which are proved and which are open.

Item	Status	Comment
Lemma 1(i): strong-coupling gap	Input [1]	Embedding certification required
$\mathfrak{G}_1$: numerical c_1, c_2	Open	Certify computable smallness bounds
$\mathfrak{E}(g, K)$ and Lemma 1(ii): area law	Conditional	Implication proved; the expansion is open
Lemma 2 and $\mathfrak{G}_2$: Hamiltonian–Euclidean bridge, reflection positivity	Proved	Fixed-volume bridge; no uniform τ limit
Lemma 3 and $\mathfrak{G}_3$: exact matching in 2D	Proved	Classical; 3D and 2+1 warm-up open
R, R′, Y, Y′, AF, AF±, $\mathfrak{G}_4$: RG through the crossover	Open	The Millennium-level step; AF measured against the two-loop scale
Lemma 4: gap $\gtrsim a(g)$	Conditional	Implication proved
Lemma 5: admissible-volume gap bounds	Conditional	Implication proved
C_1–C_4: continuum clustering, convergence, E(4) covariance, growth	Open	Full polynomial algebra required
Theorem: reconstructed observable sector	Conditional	Implication proved

The conditional RG implications organize sufficient assumptions, rather than supply new ultraviolet or continuum estimates. Lemmas 2 and 3 use established transfer-matrix and two-dimensional heat-kernel methods. The full-polynomial clustering hypothesis already contains the decisive continuum spectral information. A nontrivial Yang–Mills limit, physical scale setting, boundary control and all open RG estimates remain to be constructed; priority or novelty is not established here.

References

1. R. Cooperman and A.E. Petrunina V.V., “A Uniform Strong-Coupling Spectral Gap and a Center Selection Rule for the SU(N) Kogut–Susskind Lattice Hamiltonian,” XISOSIS, November 2026.
2. R. Cooperman and A.E. Petrunina V.V., “Toward the Weak-Coupling Continuum Limit of SU(N) Lattice Yang–Mills: Entanglement Lemmas and a Research Programme,” XISOSIS, November 2026.
3. D. A. Yarotsky, “Quasi-particles in weak perturbations of non-interacting quantum lattice systems,” arXiv:math-ph/0411042.
4. R. Cooperman and A.E. Petrunina V.V., “The Second Cooperman–Petrunina Challenge Series,” XISOSIS, Vienna, November 2026, doi:10.5281/zenodo.23975765.
5. T. Balaban, “Renormalization group approach to lattice gauge field theories I,” Comm. Math. Phys. 109 (1987).
6. H. Shen, R. Zhu and X. Zhu, “A stochastic analysis approach to lattice Yang–Mills at strong coupling,” arXiv:2204.12737.
7. J. Kogut and L. Susskind, “Hamiltonian formulation of Wilson's lattice gauge theories,” Phys. Rev. D 11 (1975) 395.
8. K. Osterwalder and E. Seiler, “Gauge field theories on a lattice,” Ann. Physics 110 (1978), 440–471.
9. T. Ichinose and H. Tamura, “Results on convergence in norm of exponential product formulas and pointwise of the corresponding integral kernels,” arXiv:0905.3458.

10. B. K. Driver, "YM_2: continuum expectations, lattice convergence, and lassos," Comm. Math. Phys. 123 (1989), 575–616.
11. A. A. Migdal, "Recursion equations in gauge field theories," Sov. Phys. JETP 42 (1975), 413–418.
12. K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," Comm. Math. Phys. 31 (1973), 83–112; II, 42 (1975), 281–305.

References [1], [2] and [4] are author-supplied companion works; their claims have not been independently verified. The two-loop coefficient b_1 is the standard universal value for pure SU(N).

Repository, citation and licence

GitHub repository: https://github.com/teknoquants/CONTINUUM-4D-SU-N-YANG-MILLS-WITH-A-MASS-GAP
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